CSL301:Operating Systems Overview

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Acknowledgement: Adopted from teaching materials Prof. Chester, IIT Madras

Administrative

- Class website: Please join google class room: zfzplksq
- Textbook:
  - *Operating System Concepts by Silberschatz, Galvin, and Gagne*
  - *Operating Systems: Three Easy Pieces by Remzi*
  - *Free online version*
- Reference operating system: xv6
- Other similar courses:
  - MIT Operating System Engineering
  - Operating Systems IITB
  - Operating Systems IITM

Administrative

- Lectures: *Mond 11:30 - 12:25, Wed 8:30-9:25, Fri 11:30- 12:25*
- Lab: Mon 3:30 – 5:20
- Grade*:*
  - *35% mid-sem*
  - *45% end-sem exam*
  - *10% lab exam*
  - *5% class tests*
  - *5% lab assignment*
  - *Bonus 10 points for show interesting OS project*

We following absolute grading

Course content

- Threads & Processes
- Scheduling
- Concurrency & Synchronization
- Virtual Memory
- Disks and File systems
- Protection & Security
- Virtual machines
- Note: Lectures will based on xv6

Course goal

- Introduce you to operating system fundamental concepts
  - Knowledge of OS concept will help you to be better system developer
- Covers widely applicable system concepts
  - Scheduling, concurrency, resource management, protection mechanism
- Teach you to deal with largest software system
  - Some programming assignment will be based on modifying entire OS

Programming assignments

- Initial assignments you will implement basic concepts on
  - Threads
  - Multiprogramming
  - Virtual memory
  - File system
- Later assignment will be to implement part of xv6 os

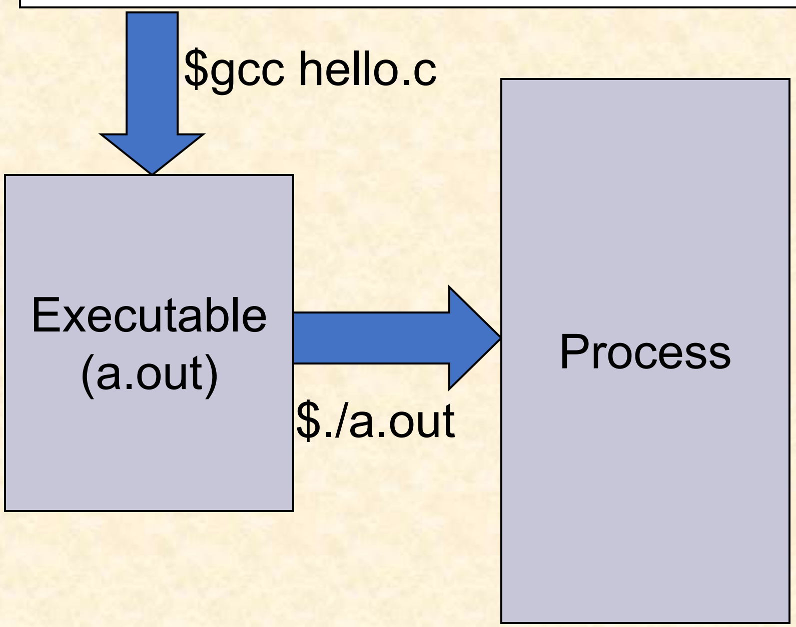
What we learn today and next class

We will take you through basic terminology of OS

Program vs process

```c
#include <stdio.h>

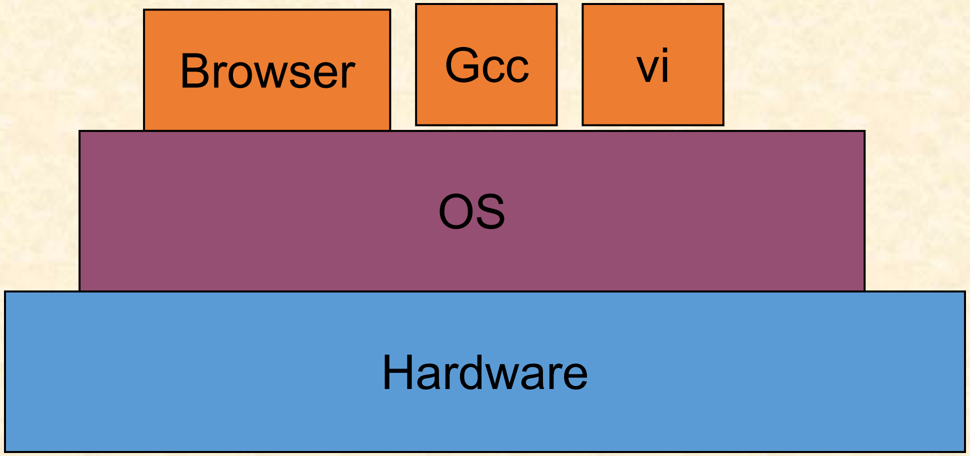
int main(){
  char str[] = "Hello World\n";
  printf("%s", str);
}
```



- Process
  - A program in execution
  - Comprises of
    - Executable instructions
    - Stack
    - Heap
    - State
  - State contains : registers, list of open files, related processes, etc.
  - Virtualization:
    - Every process think it has its own CPU, RAM, I/O

What is operating system

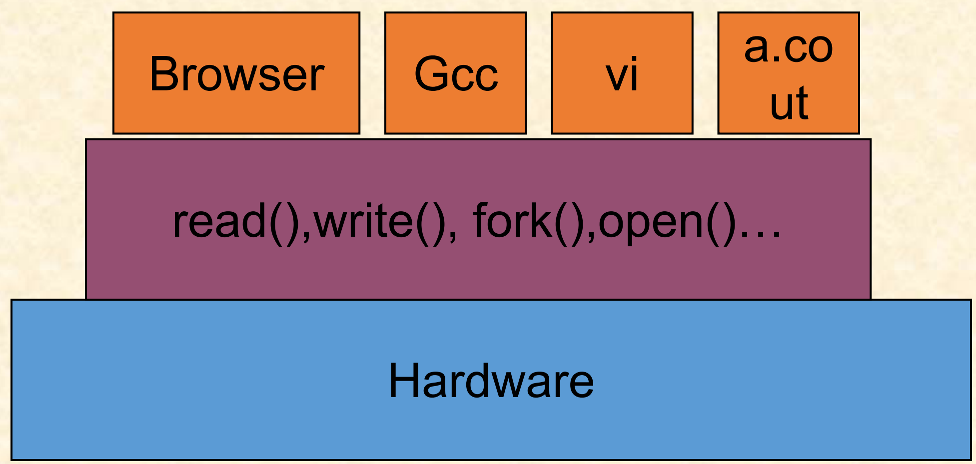
- It is a system software that control the hardware in order to facilitate the execution of application program



What is operating system

- Provides hardware abstraction to applications
  - Application program need not to know every h/w details
  - It access the hardware through OS system calls

```
int main()
{
 int fd;
char c[]="subidh";
fd=open("sk.c", O_WRONLY);
write(fd, c, 6);
close(fd);
}
```



How is open() different from fopen()???

fopen() vs open()

fopen("test.c", "r"); Library function

libc

Application Program

open(("sk.c", O_RDONLY)

kernel function /system call

File path

File mode: read/write

Implementation of Open()

Operating system

int 0x80(on older x86 systems).

What is operating system

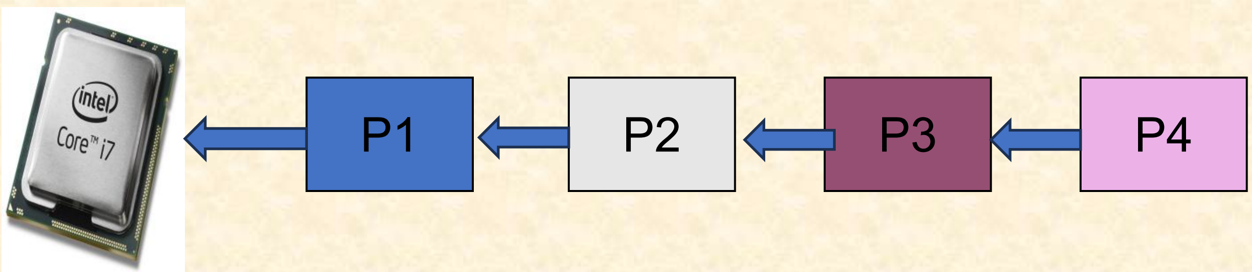
- OS is also a resource manager
  - Optimizes h/w resources for application program

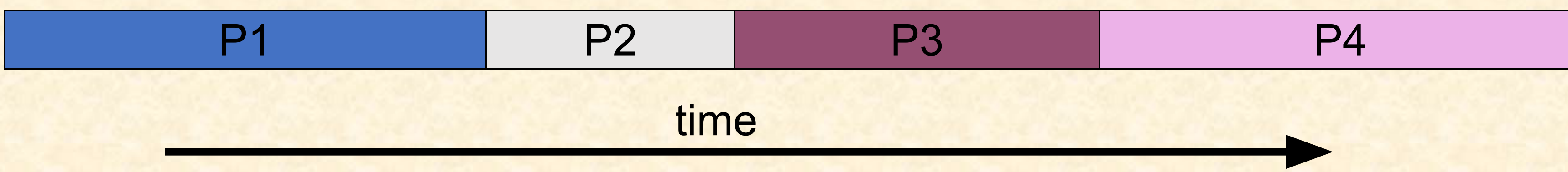

OS enables optimum way of sharing resources among processes

Sharing the CPU

Say 4 programs wants to run in CPU

Primitive approach:



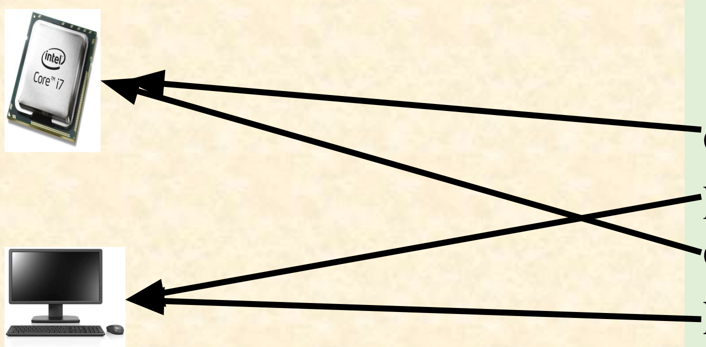


When one program completes, next starts

How can we improve this???

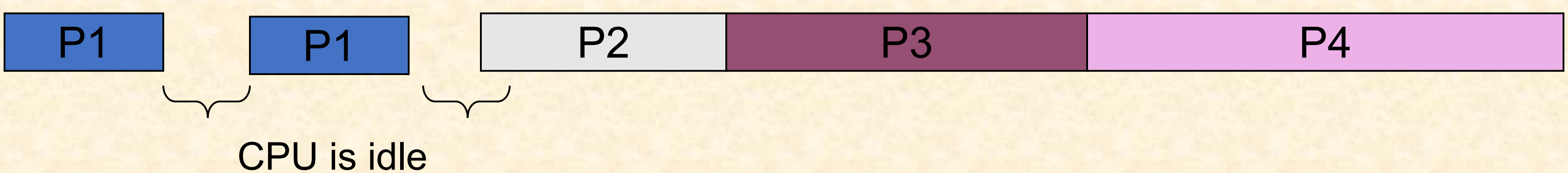
Sharing the CPU

Primitive approach:



```
int main()
{
 int a=40, b=30, c;
c=a*a+b*b;
printf("%d", c);
c=a+b;
printf("%d",c);
}
```

I/O is much slower than CPU



OS can run other program when CPU is idle

Sharing the CPU

Primitive approach:

```
int main()
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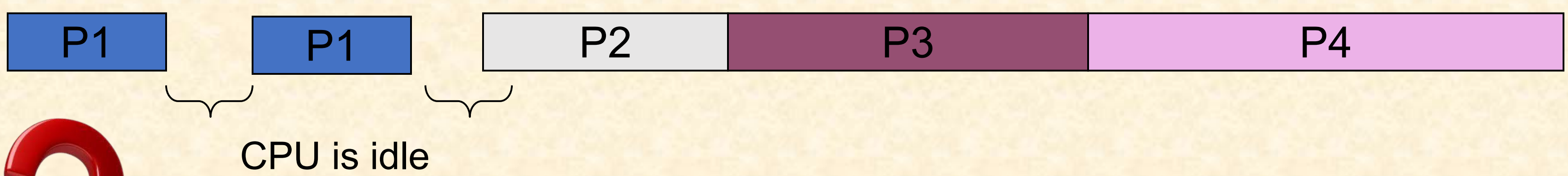
I/O is much slower than CPU



A RISC V processor speed is 1GHz. I/O speed 1MHz. Find the time it takes to execute one CPU bound instruction and one I/O bound instructions?

Multiprogramming

CPU do not sit idle

```
int main()
{
 int a=40, b=30, c;
c=a*a+b*b;
printf("%d", c);
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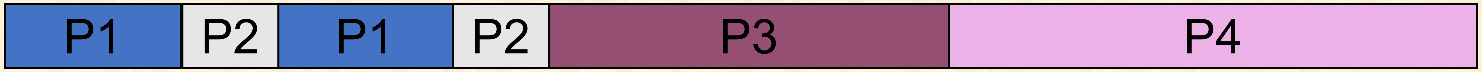
}
```

I/O is much slower than CPU



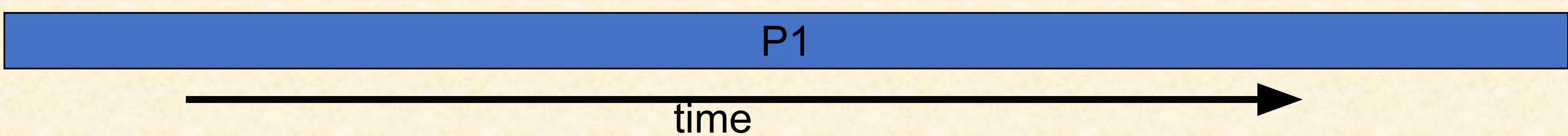
While P1 is in I/O, P2 is running in CPU

CPU utilization improves

Challenges in Multiprogramming

A program can go rogue: infinite loop

```
int main()
{
While(1)
}
```



P1 cause starvation to other programs

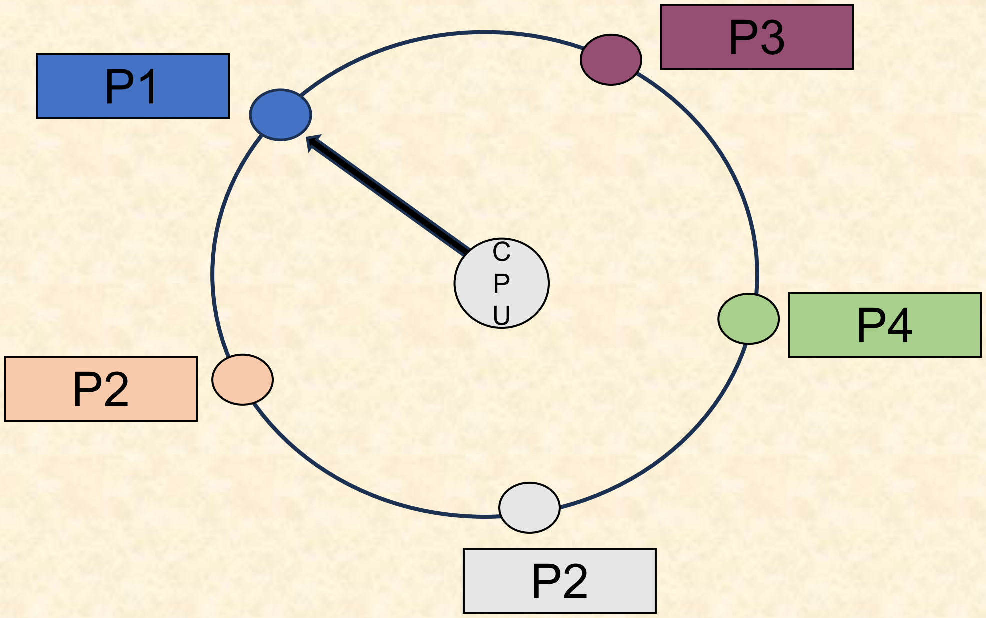
Challenges in Multiprogramming

Preemption: Forcefully removing the program

```
int main()
{
While(1)
}
```

Multitasking Operating system

- Time sliced: small but fixed time of execution for each prog.



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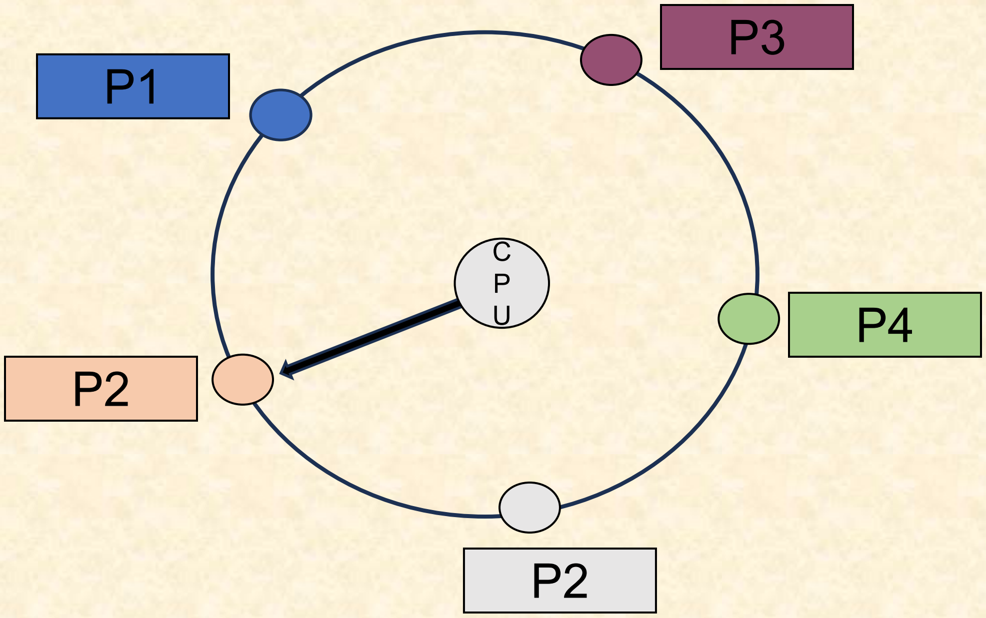

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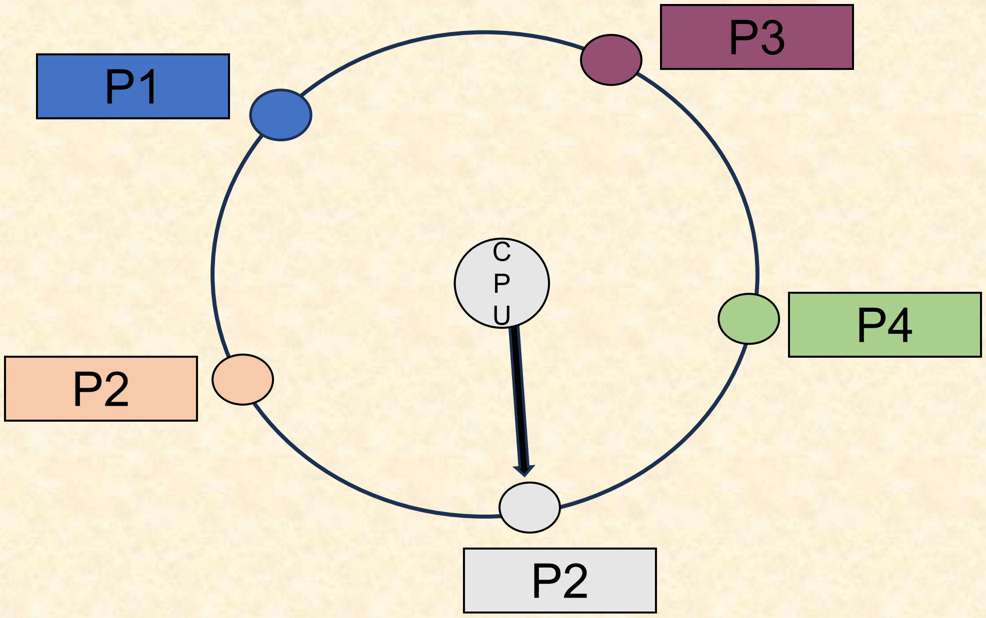

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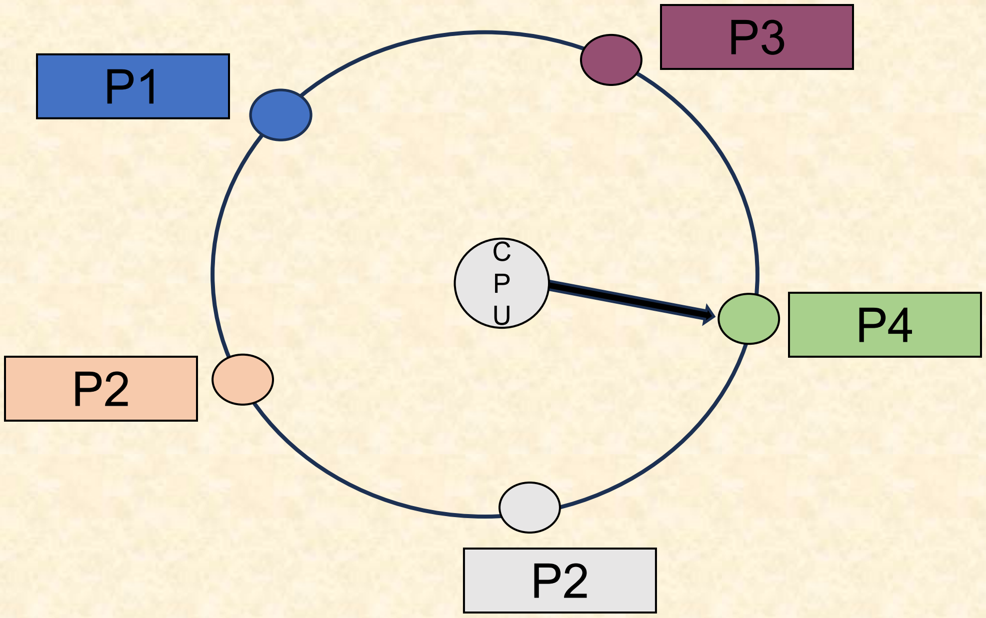
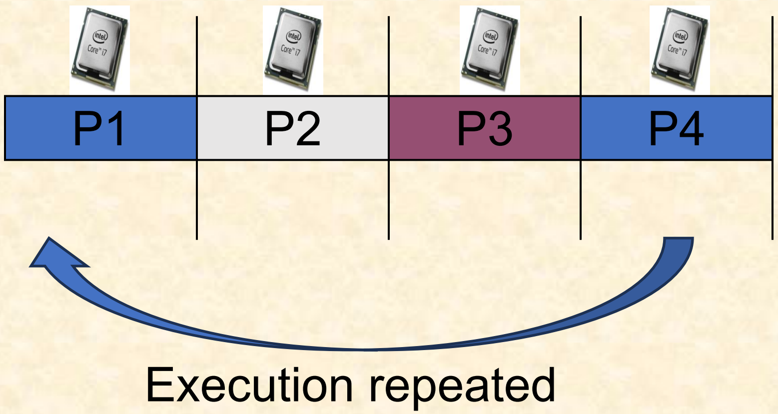


Time Sharing is virtualization of CPU

- Time sliced: small but fixed time of execution for each prog.



Virtual parallelism: Programs have impression of parallel execution

Next class OS Overview Part 2